

Assignment 2

1. (a) If the point $(a, 3)$ is at a distance of $\sqrt{5}$ units from the point $(2, a)$, find a .
(b) What will be the values of x , if the distance between $(x, -4)$ and $(-8, 2)$ is 10.
(c) If the distance between the points $(a, -5)$ and $(2, 2)$ is 13 find a .
2. Show that the four points $A(4, 2, -6)$, $B(5, -3, 1)$, $C(12, 4, 5)$ and $D(11, 9, -2)$ are the vertices of a rectangle.
3. Find the equation of the plane through the point (α, β, γ) and parallel to the plane $ax + by + cz = 0$.
4. If the angle between the planes $ax + by + cz + d = 0$ and $\alpha x + \beta y + \gamma z + \delta = 0$ is $\frac{\pi}{4}$
Then show that, $2(a\alpha + b\beta + c\gamma)^2 = (a^2 + b^2 + c^2)(\alpha^2 + \beta^2 + \gamma^2)$.